MODULE 31 · DOMAIN 5 - IMPLEMENT INFORMATION EXTRACTION SOLUTIONS

Semantic, hybrid and vector search

Vector algorithms, hybrid queries and RRF, the semantic ranker's limits, relevance tuning, agentic retrieval, and connecting retrieval to agents.

DOMAIN 5 - Information extraction WEIGHT 10-15% MODULE 31 of 39

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Retrieval has two stages: **recall**, then **ranking**. Ranking can only reorder what recall already returned.

Five ways to query an index

QUERY TYPE	HOW IT MATCHES	BEST AT
Keyword (BM25)	Term frequency in text	Exact terms, codes, names
Vector	Nearest neighbours	Meaning, synonyms
Hybrid	Both, merged by RRF	Most real applications
Hybrid plus ranker	Reranks the top 50	Relevance on prose
Agentic retrieval	Plans subqueries	Complex, multi-part questions

ON THE EXAM

Microsoft: hybrid queries with semantic ranking usually return the most relevant results.

HNSW or exhaustive KNN?

HNSW

Approximate, fast

- Graph-based approximate nearest neighbours
- Scales to large indexes
- Held in memory; uses vector index quota
- The usual production choice

EXHAUSTIVE KNN

Exact, thorough

- Brute-force scan of every vector
- Exact k nearest neighbours
- Doesn't consume vector index size quota
- Good for small sets and for testing recall

ON THE EXAM

You can build with HNSW and override with exhaustive at query time, which is Microsoft's suggested way to compare them.

Settings that shape vector results

- A **vector profile** ties a field to an algorithm and, optionally, a vectorizer.
- **k** sets how many neighbours come back. With the semantic ranker, Microsoft suggests **k = 50**.
- Only vector fields marked **searchable** are searched.
- **Quantization** compresses vectors to cut index size, with optional rescoring to recover quality.
- Vectors themselves can't be boosted with scoring profiles: relevance tuning works on text fields.

> ON THE EXAM
>
> Return text fields with select. A raw vector in the results helps nobody.

One request, two searches, one ranking

- One request carries both **search** text and **vectorQueries**. They run **in parallel**.
- **Reciprocal Rank Fusion** merges the two rankings into a single result set.
- Text features still work on text fields: **filters, facets, sorting, scoring profiles**.
- **Vector weighting** raises or lowers the vector side's influence.
- A **debug** parameter returns subscores, showing what each side contributed.

> ON THE EXAM
>
> RRF merges by rank position, not by raw score, so the two scoring scales never need to be comparable.

Vectors find **meaning**. Keywords find **exact codes**. Hybrid gets both, and filters still work on the text side.

What reranking does, and doesn't

- Reranks the **top 50** results from first-stage ranking, using language models.
- Returns **captions**, and optionally **answers**, extracted **verbatim** from your content.
- It **can't invent text**. If no answer-like passage exists, no answer is returned.
- Needs a **semantic configuration**: one title field, plus content and keyword fields in priority order.
- It can be added to an existing index **without rebuilding** it.

> ON THE EXAM
>
> Query rewrite (preview) expands a query into similar phrasings before retrieval, which lifts recall as well as ranking.

The ranker reranks the top 50. If the right document never reached those 50, reranking can't rescue it.

A hybrid query with semantic ranking

```
{
  "search": "cancel an order after 30 days",
  "vectorQueries": [
    {"kind": "text", "text": "cancel an order after 30 days",
     "fields": "contentVector", "k": 50}],
  "queryType": "semantic",
  "semanticConfiguration": "default",
  "filter": "region eq 'UK'",
  "select": "title,content,url",
  "top": 10
}
```

ON THE EXAM

Keyword and vector recall, filtered to the UK, reranked semantically, returning ten readable results.

When results are close but wrong

- **Scoring profiles** boost fields, freshness or tags on the text side.
- **Vector weighting** shifts the balance between meaning and keywords.
- **Filters** remove whole categories that should never match.
- **Debug subscores** show whether the text or vector side put a result where it is.
- Measure changes with the **Document Retrieval** evaluator from module 15.

> ON THE EXAM
>
> Change one setting at a time, and re-measure. Retrieval tuning without measurement is guesswork.

When one query isn't enough

- A **knowledge base** plus at least one **knowledge source** is required.
- A model **plans subqueries**, which run **simultaneously** as keyword, vector or hybrid searches.
- **Each subquery is semantically reranked**, and references are kept for citations.
- **Merged content is always returned.** References and the activity log are optional.
- More model calls and more searches mean **more latency and cost** than a single query.

> ON THE EXAM
>
> Your application calls the knowledge base, then passes the grounding data to a model, as module 13 showed.

Single query or agentic retrieval?

SINGLE HYBRID QUERY

One question, one search

- Lowest latency and cost
- Predictable behaviour
- Good for focused questions

AGENTIC RETRIEVAL

Planned subqueries

- Splits compound and comparative questions
- Uses conversation context
- Returns citations and an activity log
- Costs more, takes longer

ON THE EXAM

'Compare X and Y', or a follow-up that depends on earlier turns, is where planning earns its cost.

From index to answer

- Attach the **Azure AI Search tool** to an agent, with a connection, an index and a query type.
- Or attach a **knowledge base**, which agents call through its MCP retrieve tool.
- Defaults from module 18: **vector_semantic_hybrid** query type, **top_k** of 5.
- Permission metadata from ingestion filters results per user, as module 29 showed.
- Index a **retrievable URL** field so answers can cite their sources.

> ON THE EXAM
>
> The retrieval decisions in this module become an agent's tool configuration in module 18's terms.

Retrieval symptoms and fixes

SYMPTOM	FIX
Exact part numbers don't match	Add keyword search: go hybrid
Synonyms and paraphrases miss	Add vectors: go hybrid
Right document exists but ranks low	Improve recall, then add the semantic ranker
No semantic answers returned	No answer-like text in the results; the ranker can't invent it
Compound questions half-answered	Agentic retrieval with subqueries
Users see documents they shouldn't	Permission metadata and filters, from ingestion

ON THE EXAM

Ask which stage failed: recall, ranking, or the question itself needing decomposition.

> A policy search uses keyword search with the semantic ranker enabled. A policy that clearly answers a common question ranks about 180th by keyword score, and never appears.
>
> Adding the semantic ranker made no difference for this question.

What should the team do?

A Raise top to 200 so the ranker has more results to rerank.

B Add more content fields to the semantic configuration's priority list.

C Add vectors and query hybrid with k set to 50, then rerank.

D Lower the reranker score threshold the application applies to results.

◆ PAUSE AND DECIDE

What should the team do?

A Raise top to 200.

Only the top 50 ever reach the ranker, whatever top is set to.

B More content fields.

Helps the ranker judge documents it receives. This one never arrives.

C Hybrid with vectors, k = 50.

Vector recall surfaces the policy by meaning, so reranking can promote it.

D Lower threshold.

Filters the ranked list, which still doesn't contain the document.

THE SIGNAL

Ranked 180th and invisible: a recall problem, not a ranking problem.

> A team of 20,000 documents must prove, for an audit, that their vector search returns the true nearest neighbours for a fixed test set.
>
> The index currently uses HNSW, and results vary slightly between runs of the same test.

What should the team use for the audit test?

A HNSW with a much higher search-time parameter to approach exact results.

B Exhaustive KNN, which returns the exact k nearest neighbours.

C The semantic ranker, which reorders results deterministically.

D A scoring profile that boosts the fields used in the test set.

◆ PAUSE AND DECIDE

What should the team use for the audit test?

A Tuned HNSW.
Closer to exact, but still approximate by design.

B Exhaustive KNN.
Brute-force scan returns exact neighbours, and doesn't use vector index quota.

C Semantic ranker.
Reorders candidates. It doesn't make nearest-neighbour search exact.

D Scoring profile.
Tunes text relevance, and can't boost vector fields at all.

THE TIEBREAKER

'Prove exact nearest neighbours' is exhaustive KNN, and it can be an override at query time.

> A retailer's site search must match both exact SKUs, like AB-4471, and natural phrases like 'waterproof coat for hiking'.
>
> Shoppers also filter by category and by store region, and results should be as relevant as possible.

What query design fits?

A Vector search alone, since embeddings capture product meaning well.

B Keyword search alone, since SKUs and categories are exact strings.

C Vector search with a scoring profile boosting the SKU field heavily.

D Hybrid search with filters, and the semantic ranker on top.

◆ PAUSE AND DECIDE

What query design fits?

A. Vector only.
Embeddings often miss exact codes like AB-4471.

B. Keyword only.
Misses 'waterproof coat for hiking' when the listing says 'rain jacket'.

C. Vector plus scoring profile.
Scoring profiles tune text relevance; they can't rescue vector recall of a code.

D. Hybrid, filters, semantic ranker.
Keywords catch codes, vectors catch meaning, filters and ranking still apply.

THE SIGNAL

Exact codes and natural phrases in the same box is the textbook case for hybrid.

> A support copilot is asked things like 'How does the returns policy differ between the UK and Germany after 30 days?'. Single hybrid queries answer one part and miss the other.
>
> The team also needs citations, and a record of what was searched, for review.

What should the team use?

A Agentic retrieval over a knowledge base, with citations.

B A larger top value, so more documents reach the model in one query.

C Bigger chunks with more overlap, so each chunk covers more ground.

D A fine-tuned model that has learned both countries' returns policies.

◆ PAUSE AND DECIDE

What should the team use?

A Agentic retrieval.

Plans subqueries per part, reranks each, and returns references plus an activity log.

B Larger top.

More results from one query that still doesn't ask about both countries.

C Bigger chunks.

Dilutes relevance, and UK and German policies live in different documents anyway.

D Fine-tuning.

Behaviour, not facts, and it removes the citations the team needs.

THE SIGNAL

Comparative questions plus a need for citations and an audit trail: agentic retrieval.

Querying in five lines

- **Recall first, ranking second.** Ranking can't rescue what recall missed.
- **HNSW** for scale, **exhaustive KNN** for exact results and testing.
- **Hybrid plus semantic ranking** is the strong default; filters still apply.
- **The ranker sees only the top 50**, and extracts captions and answers verbatim.
- **Agentic retrieval** for compound questions, with references and an activity log.

> ON THE EXAM
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> Next, module 32: Content Understanding and document extraction, closing Domain 5's pipeline work.